AAT REPORT ON

Design and Verification of Combinational and Sequential Digital Modules using Verilog HDL

COURSE: ADVANCE DIGITAL LOGIC DESIGN

(23EC5PE1AD)

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Academic Year

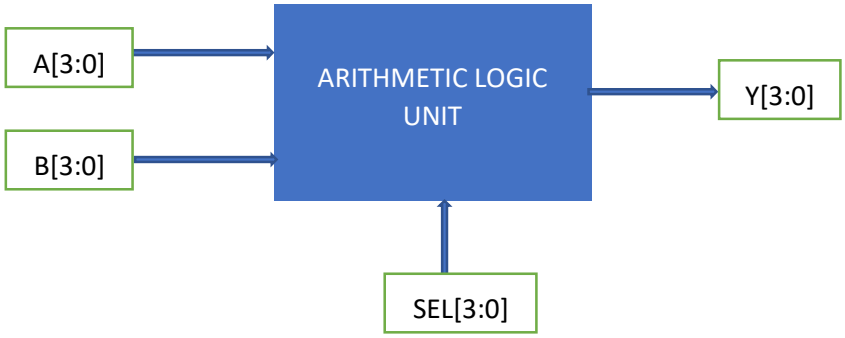
2026 – 2027

VERILOG CODES

1. ARITHMETIC LOGIC UNIT (ALU)

An ALU is a combinational circuit that performs arithmetic and logical operations on binary data. It performs operations such as addition, subtraction, AND, OR, and XOR. The operation is selected using control or select lines. The output depends on the input data and the selected operation.

DESIGN



VERILOG CODE

```
module alu (     input
[3:0] A, B,     input
[2:0] sel,     output
reg [3:0] Y

);
always @(*) begin
    case (sel)
        3'b000: Y = A + B;
        3'b001: Y = A - B;
        3'b010: Y = A & B;
        3'b011: Y = A | B;
3'b100: Y = A ^ B;
default: Y = 4'b0000;
endcase end
endmodule
```

Test bench: module tb alu; reg [3:0] A,

B; reg [2:0] sel; wire [3:0] Y; alu uut

(.A(A), .B(B), .sel(sel), .Y(Y)); initial

begin A=4'd5; B=4'd3;

sel=3'b000; #10; sel=3'b001; #10;

sel=3'b010; #10; sel=3'b011; #10;

sel=3'b100; #10;

$finish; end

endmodule

2. COMPARATOR

A comparator is a combinational circuit used to compare two binary numbers. It determines whether the first number is greater than, less than, or equal to the second number. It produces three outputs: greater than, less than, and equal. The outputs depend on the values of the input numbers.

DESIGN



VERILOG CODE

module comparator (

input [3:0] A, B,

output gt, lt, eq

);

assign gt = (A > B);

assign lt = (A < B);

```
assign eq = (A == B);
endmodule
```

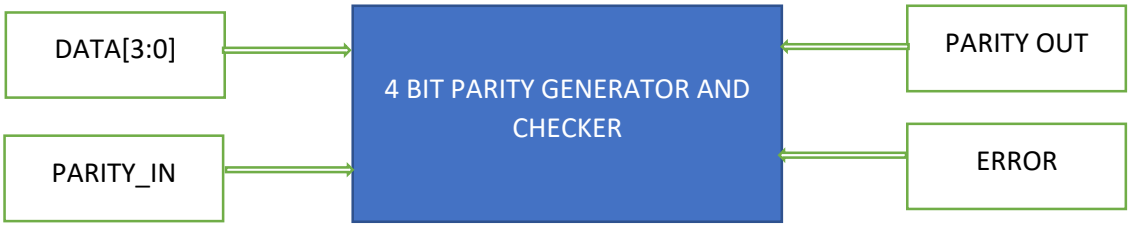
TEST BENCH:

```
module tb_comparator; reg [3:0] A, B; wire gt, lt, eq;
comparator uut (.A(A), .B(B), .gt(gt), .lt(lt), .eq(eq));
initial begin     A=5; B=3; #10;
   A=2; B=7; #10;
   A=4; B=4; #10;
   $finish; end
endmodule
```

3. 4-BIT PARITY GENERATOR AND CHECKER

A parity generator generates a parity bit to detect errors in binary data. It uses XOR gates to generate even or odd parity for a 4-bit input. A parity checker checks the received data along with its parity bit. If the parity is incorrect, an error is detected**.**

DESIGN



VERILOG CODE

```
module parity (
input [3:0] data,
```

```
input parity_in,
output parity_out,
output error
);
assign parity_out = ^data; assign
error = ^{data, parity_in};
endmodule
```

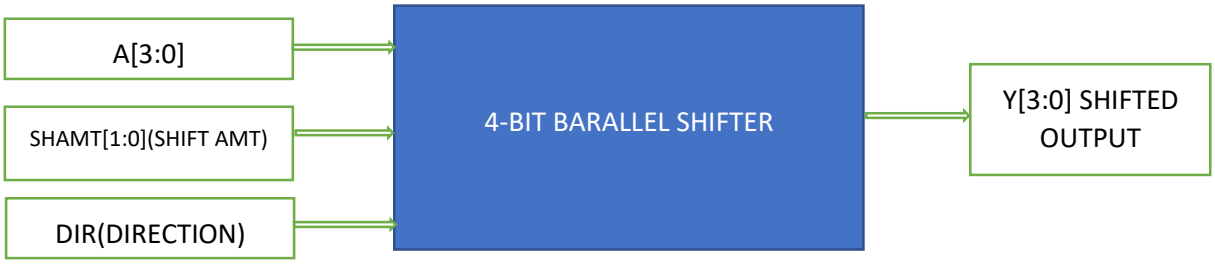
TEST BENCH:

```
module tb_parity; reg
[3:0] data; reg
parity_in; wire
parity_out, error;
parity uut (.data(data), .parity_in(parity_in), .parity_out(parity_out), .error(error)); initial
begin
   data=4'b1010; parity_in=1'b0; #10;
parity_in=1'b1; #10;     data=4'b1110;
parity_in=1'b0; #10;     $finish; end
endmodule
```

4. 4-BIT BARALLEL SHIFTER

A barrel shifter is a combinational circuit used to shift binary data by a specified number of positions. It can perform left or right shifts depending on the control input. The shift amount is selected using control lines. It performs shifting in a single operation without requiring multiple clock cycles.

DESIGN



VERILOG CODE

```
module barrel_shifter (
input [3:0] A,     input
[1:0] shamt,     input
dir,     output reg [3:0]
Y

);
always @(*) begin
if (dir == 1'b0)         Y
= A << shamt;     else

      Y = A >> shamt;
end
endmodule
```

TEST BENCH:

```
module tb_barrel;
reg [3:0] A; reg
[1:0] shamt; reg
dir; wire [3:0] Y;

barrel_shifter uut (.A(A), .shamt(shamt), .dir(dir), .Y(Y)); initial
begin
```

```
    A=4'b1010; shamt=1; dir=0; #10;
shamt=2; dir=1; #10;     shamt=0;
dir=0; #10;

    $finish; end
endmodule
```

5. TIMER MODULE

A timer module counts clock pulses to measure a specific time interval. It uses a counter that increments with each active clock edge. When the counter reaches the specified value, the timer generates an output signal. A reset signal initializes the counter to zero.

DESIGN



VERILOG CODE

```
module timer (
    input clk, reset,
output reg done

);
reg [3:0] count;
always @(posedge clk or posedge reset) begin
if (reset) begin         count <= 0;         done <=
0;     end     else if (count == 9) begin
count <= 0;         done <= 1;     end     else
```

```
begin        count <= count + 1;        done <=
0;     end end endmodule
```

TEST BENCH:

```
module tb_timer;
reg clk, reset;
wire done;

timer uut (.clk(clk), .reset(reset), .done(done));
always #5 clk = ~clk;
initial begin
clk=0; reset=1;
#10 reset=0;
#120 $finish; end
endmodule
```

6. RING COUNTER

A ring counter is a sequential circuit made using a shift register with its output connected back to its input. It circulates a single logic 1 through the register during each clock pulse. A reset initializes the counter to a known state. For a 4-bit ring counter, the sequence normally contains four distinct states.

DESIGN



VERILOG CODE

```
module ring_counter (
input clk, reset,
output reg [3:0] q

);
always @(posedge clk or posedge reset) begin
if (reset)        q <= 4'b0001;     else
      q <= {q[2:0], q[3]};
end endmodule
```

TEST BENCH:

```
module tb_ring;
reg clk, reset;
wire [3:0] q;

ring_counter uut (.clk(clk), .reset(reset), .q(q));
always #5 clk = ~clk; initial begin     clk=0;
reset=1;     #10 reset=0;     #80 $finish; end
endmodule
```

7. JOHNSON COUNTER

A Johnson counter is a sequential circuit in which the inverted output of the last flip-flop is connected to the input of the first flip-flop. register shifts its data on every active clock edge. An n-bit Johnson counter can generate up to 2n distinct states. It is used in sequence generation and digital control circuits.

DESIGN



VERILOG CODE

```
module johnson_counter (
   input clk, reset,
output reg [3:0] q

);
always @(posedge clk or posedge reset) begin
   if (reset)        q <=
4'b0000;     else        q <=
{q[2:0], ~q[3]}; end
endmodule
```

TEST BENCH

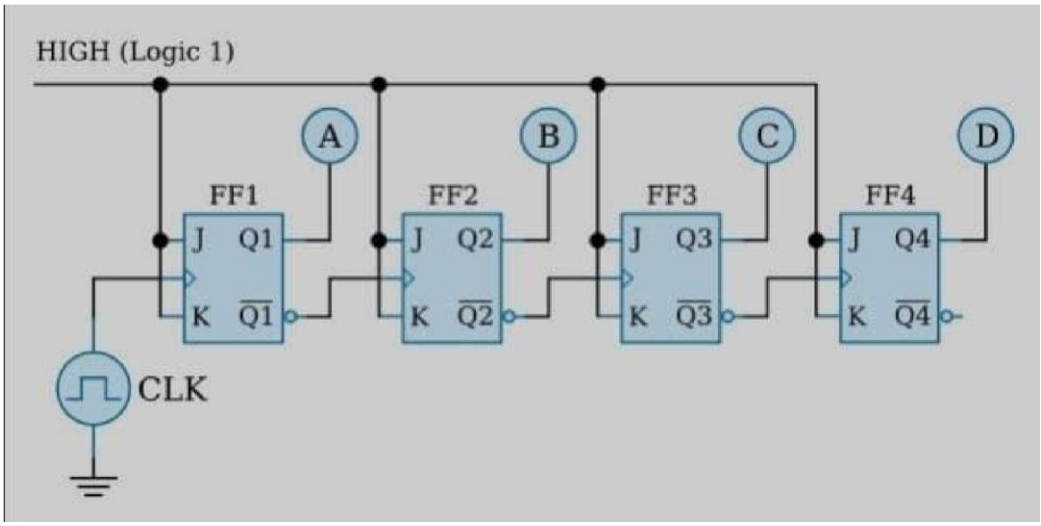
```
module tb_johnson;
reg clk, reset; wire
[3:0] q;

johnson_counter uut (.clk(clk), .reset(reset), .q(q));
always #5 clk = ~clk; initial begin     clk=0;
reset=1;     #10 reset=0;     #100 $finish; end
endmodule
```

8. SYNCHRONOUS COUNTER

In a synchronous counter, all flip-flops receive the same clock signal. All flip-flops change their states together at the active clock edge. It provides faster operation because the flip-flops are triggered simultaneously.

DESIGN



VERILOG CODE

```
module sync_counter (
   input clk, reset,
output reg [3:0] q

);
always @(posedge clk or posedge reset) begin
if (reset)        q <= 4'b0000;     else

      q <= q + 1'b1;
end endmodule
```

TEST BENCH:

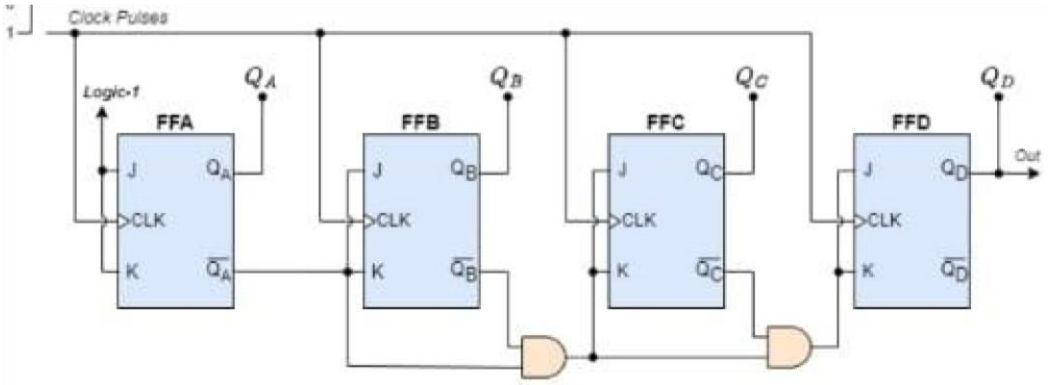
```
module tb_sync;
reg clk, reset; wire
[3:0] q;

sync_counter uut (.clk(clk), .reset(reset), .q(q));
always #5 clk = ~clk; initial begin     clk=0;
reset=1;     #10 reset=0;     #180 $finish; end
endmodule
```

9. ASYNCHRONOUS COUNTER

In an asynchronous counter, only the first flip-flop receives the external clock directly. The output of one flip-flop acts as the clock for the next flip-flop. The changes propagate through the counter, causing cumulative propagation delay.

DESIGN



VERILOG CODE

```
module async_counter (
   input clk, reset,
output reg [3:0] q

);
```

```
always @(posedge clk or posedge reset) begin
if (reset) q[0] <= 0;     else       q[0] <= ~q[0];
end

  always @(negedge q[0] or posedge reset) begin
if (reset) q[1] <= 0;     else       q[1] <= ~q[1]; end

always @(negedge q[1] or posedge reset) begin
if (reset) q[2] <= 0;     else       q[2] <= ~q[2];
end

always @(negedge q[2] or posedge reset) begin
if (reset) q[3] <= 0;     else       q[3] <= ~q[3];
end endmodule
```

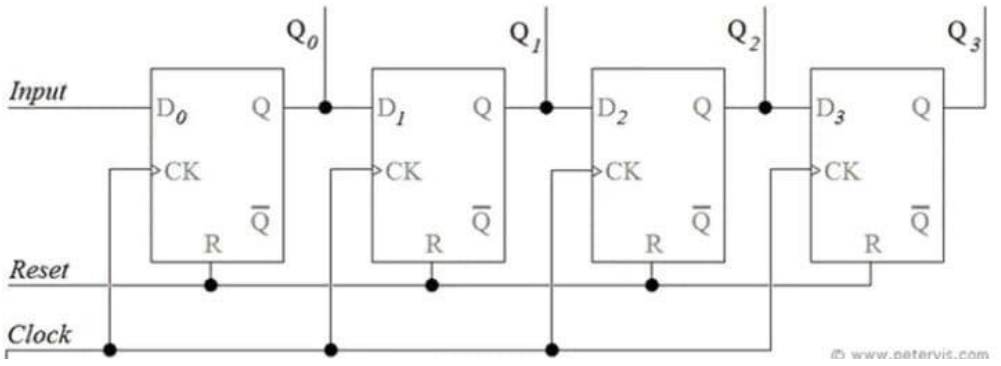
TEST BENCH:

```
module tb_async;
reg clk, reset; wire
[3:0] q;
async_counter uut
(.clk(clk),
.reset(reset),
.q(q)); always #5
clk = ~clk; initial
begin     clk=0;
reset=1;     #10
reset=0;     #180
$finish; end
endmodule
```

10.SHIFT REGISTER

A shift register is a sequential circuit used to store and shift binary data. It consists of a group of flip-flops connected in series. At every active clock edge, data shifts from one flip-flop to the next. It is used for temporary data storage and serial-to-parallel or parallel-to-serial conversion.

DESIGN



VERILOG CODE

```
module shift_register (
input clk, reset, serial_in,
output reg [3:0] q

);
always @(posedge clk or posedge reset) begin
if (reset)        q <= 4'b0000;     else

      q <= {q[2:0], serial_in};
end endmodule
```

TEST BENCH:

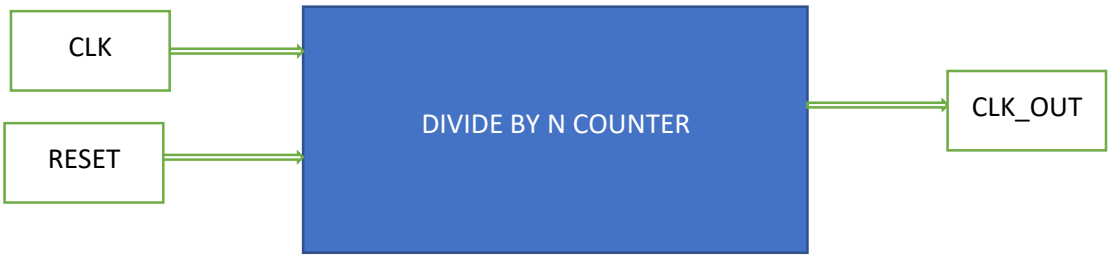
```
module tb_shift; reg
clk, reset, serial_in;
wire [3:0] q;
```

```
shift_register uut ( .clk(clk), .reset(reset), .serial_in(serial_in), .q(q));
always #5 clk = ~clk; initial begin
    clk=0; reset=1; serial_in=0;
    #10 reset=0;
serial_in=1; #10;
serial_in=0; #10;
serial_in=1; #10;
serial_in=1; #10;
#30 $finish; end
endmodule
```

11.DIVIDE BY N COUNTER

A divide-by-N counter divides the input clock frequency by a specified number N. It counts clock pulses from zero to N−1 and then resets to zero. A suitable output signal is generated based on the counter state. It is used in clock division and frequency control circuits.

DESIGN



VERILOG CODE

```
module divide_by_n (
    input clk, reset,
output reg clk_out
);
```

```
reg [3:0] count;
always @(posedge clk or posedge reset) begin
if (reset) begin        count <= 0;        clk_out
<= 0;     end     else if (count == 9) begin
count <= 0;         clk_out <= ~clk_out;     end
else begin         count <= count + 1'b1;     end
end
endmodule
```

TEST BENCH:

```
module tb_divide;
reg clk, reset; wire
clk_out;

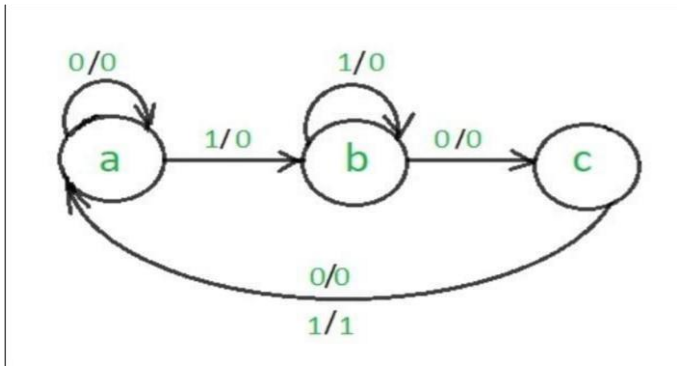
divide_by_n uut (.clk(clk), .reset(reset), .clk_out(clk_out));
always #5 clk = ~clk; initial begin     clk=0; reset=1;     #10
reset=0;     #250 $finish; end endmodule
```

12. MELAY FSM

A Mealy finite state machine produces its output based on the present state and current input. Its output can change immediately when the input changes. It generally requires fewer states than an equivalent Moore machine. It is used in sequence detectors and digital control systems.

This example detects the input sequence 101. The output becomes 1 when the final 1 completes the sequence, and the output depends on both the current state and input.

DESIGN



VERILOG CODE

```
module mealy_fsm (
   input clk, reset, din,
output reg detected

); reg [1:0] state; local param S0=0, S1=1,
S2=2; always @(posedge clk or posedge reset)
begin     if (reset) begin         state <= S0;
detected <= 0;     end     else begin
detected <= 0;         case (state)            S0: if
(din) state <= S1;                else state <= S0;
S1: if (din) state <= S1;                else state <=
S2;                    S2: if (din) begin
state <= S1;                    detected <= 1;
end               else state <= S0;  default: state
<= S0;         endcase
  end end
endmodule
```

TEST BENCH

```
module tb_mealy; reg
clk, reset, din; wire
detected;
```

```
mealy_fsm uut (.clk(clk), .reset(reset), .din(din), .detected(detected));

always #5 clk = ~clk; initial begin     clk=0; reset=1; din=0;

   #10 reset=0;

din=1; #10;     din=0;

#10;     din=1; #10;

din=0; #10;     din=1;

#10;     din=0; #10;

   $finish;

end

endmodule
```

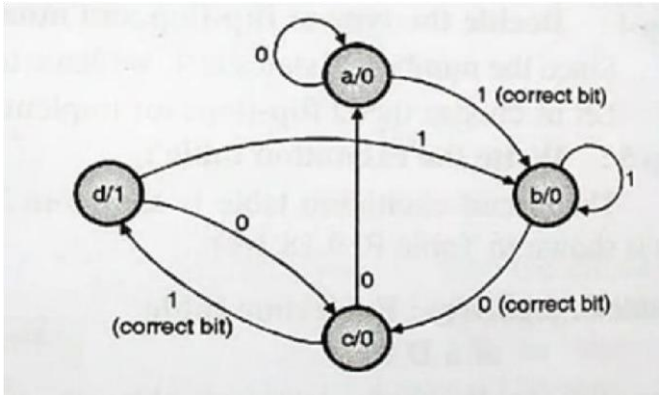
13.MOORE FSM

A Moore finite state machine produces its output based only on the present state. Its output changes when the machine transitions to another state. The output is generally more stable because it does not depend directly on the current input. It is used in controllers and sequence detection circuits.

This example also detects the sequence 101. The output becomes 1 in a dedicated detection state, so it depends only on the current state

.

DESIGN



VERILOG CODE

```
module moore_fsm (
   input clk, reset, din,
output detected

); reg [1:0]
state;

localparam S0=0, S1=1, S2=2, S3=3; assign
detected = (state == S3); always @(posedge
clk or posedge reset) begin     if (reset)
state <= S0;     else begin         case (state)

          S0: if (din) state <= S1;
else state <= S0;           S1: if
(din) state <= S1;
else state <= S2;          S2: if
(din) state <= S3;
else state <= S0;
        S3: if (din) state <= S1;
else state <= S2;

      default: state <= S0;
endcase

end

end

endmodule
```

TEST BENCH:

```
module tb_moore;
```

```
reg clk, reset, din; wire
detected;

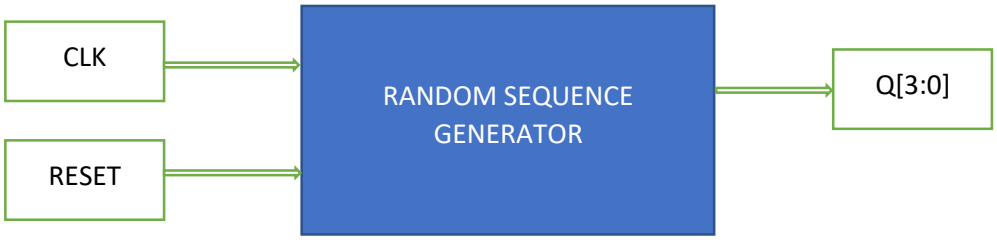
moore_fsm uut (.clk(clk), .reset(reset), .din(din), .detected(detected));
always #5 clk = ~clk; initial begin     clk=0; reset=1; din=0;     #10
reset=0;     din=1; #10;     din=0; #10;     din=1; #10;     din=0; #10;
din=1; #10;

   #20 $finish;
end
endmodule
```

14.RANDOM SEQUENCE GENERATOR

A random sequence generator produces a sequence of binary values that appears random. It can be implemented using a Linear Feedback Shift Register (LFSR). The circuit uses shift registers and XOR feedback to generate different bit patterns. It is commonly used in testing digital circuits and generating pseudorandom sequences.

DESIGN



VERILOG CODE

```
module lfsr (
input clk, reset,
output reg [3:0] q

);
always @(posedge clk or posedge reset) begin
if (reset)        q <= 4'b0001;     else

      q <= {q[2:0], q[3] ^ q[2]};
end

endmodule
```

TEST BENCH:

```
module tb_lfsr;
reg clk, reset; wire [3:0] q; lfsr uut
(.clk(clk), .reset(reset), .q(q)); always
#5 clk = ~clk; initial begin     clk=0;
reset=1;     #10 reset=0;     #200
$finish;

 end

endmodule
```